Statistical Assurance Regions for Predictive Models Under Joint Missingness and Distribution Shift

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Abstract

A frozen predictive pipeline can satisfy a risk requirement in some deployment conditions and fail in others. We study statistical assurance regions (SARs), estimated inner subsets of prespecified environment families whose predictive risks are simultaneously bounded by a tolerance. The SciGuard workflow combines finite-sample exact-binomial and bounded-loss procedures with a dependence-aware Gaussian multiplier band, explicit sampling constructions, reference-risk accounting, and independent-pilot design. Calibration shows that the multiplier can gain efficiency but is not a default finite-sample method for binary loss: simultaneous coverage was 0.928 in the primary design and 0.780 in a rare-event design, versus nominal 0.95. Experiments include nested refits, missingness mechanisms, WDBC stress tests, and credentialed MIMIC-IV v3.1 evaluation on 35,114 deployment admissions. MIMIC domain labels are approximate and calendar-overlapping because of patient-specific date shifting; we do not claim measured covariate or outcome shift across them. Under a permissive 0–1-loss tolerance, all 60 eligible conditions, including an always-survive baseline, were certified. Under a secondary false-negative-risk target requiring sensitivity at least 0.80, exact Bonferroni certification retained only 3 of 60 conditions, all for histogram gradient boosting. SAR certification is therefore relative to the chosen loss, tolerance, environment family, and sampling contract, not a claim of clinical safety.

Keywords: deployment assurance, distribution shift, missing data, simultaneous inference, risk control

1 Introduction

An acceptable test-set average does not specify the conditions under which a trained model can meet a performance requirement. A hospital, for example, may encounter changes in both its patient population and the availability of recorded measurements. A useful validation report would identify which explicitly modeled combinations remain supported by the data, while distinguishing an observed low error rate from a statistically bounded risk.

We study this reporting problem for a frozen predictive pipeline. Given environments $e_1, \ldots, e_K$, predictive risks $R_1, \ldots, R_K$, and a tolerance τ, the target is $\mathcal{A}_\tau = \{e_k : R_k \leq \tau\}$. A simultaneous upper band $(U_1, \ldots, U_K)$ produces the certified inner estimate $\widehat{\mathcal{A}}_\tau = \{e_k : U_k \leq \tau\}$. For example, upper bounds $(.07, .09, .13)$ at tolerance .10 certify the first two environments. The third remains uncertified even if its point estimate is below .10; it has

not thereby been proven unsafe. This distinction is the basis of the statistical assurance region (SAR).

The inferential devices underlying this construction are established. Exact binomial limits, bounded-sum inequalities, multiple testing, and Gaussian multiplier approximation already provide ways to control joint error (Clopper and Pearson, 1934; Hoeffding, 1963; Holm, 1979; Chernozhukov et al., 2013). Learn then Test (LTT) can directly index its hypotheses by deployment environments (Angelopoulos et al., 2025). Merely naming the resulting set does not create a new inferential theorem. We therefore study SARs as an applied methodology and implement them in the SciGuard software workflow, which connects an explicit environment design, sampling assumptions, uncertainty accounting, and an interpretable set-valued report.

The paper makes five substantive contributions to that workflow. First, it separates finite-sample risk control from asymptotic dependence-aware inference and states when each is appropriate. Second, it gives the row-level construction needed to justify reuse of observations under multiple transformations, including corrections to a missingness generator that previously did not implement the claimed MAR mechanism. Third, it puts nested fitted-model experiments and reference-risk uncertainty at the center of the evaluation. Fourth, it provides one executable notebook and shared software implementation from which all current tables and figures are generated. Fifth, it executes a credentialed MIMIC-IV v3.1 evaluation using admission-aligned approximate temporal domains, one admission per patient, native and added missingness accounting, two learned models, and an always-survive baseline. The permissive primary 0–1-loss specification is complemented by a secondary class-sensitive false-negative-risk specification requiring sensitivity of at least 0.80. The MIMIC analysis is deliberately interpreted as an assurance-workflow benchmark rather than evidence of measured temporal shift or clinical safety.

The results do not support universal use of the multiplier method. It can enlarge the certified region when losses across environments are strongly dependent, but its simultaneous coverage was 0.928 in the primary calibration (95% Monte Carlo interval $[0.902, 0.948]$) and fell to 0.780 in the rare-event design, both below the nominal 0.95 target. We therefore use exact Clopper–Pearson Bonferroni or exact Holm as the default certification routes for binary losses; the multiplier is a dependence-aware secondary route whose finite-sample approximation must be validated rather than assumed.

2 Relationship to Existing Work

Risk control and simultaneous inference. Risk-controlling prediction sets and LTT provide model-agnostic calibration and testing frameworks (Bates et al., 2021; Angelopoulos et al., 2025). In our finite-family setting, the hypotheses $H_{0k} : R_k \geq \tau$ are an immediate LTT application. The environment-indexed assurance report is an operational interpretation of this framework, not a disjoint class of guarantees. Simultaneous confidence bands also have a substantial history beyond high-dimensional Gaussian maxima; for example, Degras (2011) studies bands for functional regression. Here the coordinates are prespecified environment-specific predictive risks. Their construction and sampling design, rather than a new general band theorem, are the focus.

Prediction and adaptation under shift. ATC predicts target accuracy from source-calibrated confidence scores and unlabeled target observations (Garg et al., 2022). Its inputs and objective differ from labeled environment certification. We therefore evaluate it using risk-prediction error, not by ranking its "certification" error against a formal familywise procedure. Weighted conformal prediction addresses covariate shift using likelihood-ratio information under its stated conditions (Tibshirani et al., 2019); it does not by itself give our simultaneous region over an arbitrary family of missingness interventions. Missingness-shift adaptation changes a predictor for target conditions (Zhou et al., 2023). An adapted pipeline could subsequently be frozen and assessed with the same assurance workflow.

Generalization bounds and robust learning. PAC-Bayes analysis controls the generalization of distributions over predictors using complexity terms (Alquier, 2024). Distributionally robust optimization changes the training objective to improve performance over an uncertainty set (Duchi and Namkoong, 2021). Our main route instead conditions on a fitted pipeline and uses independent labeled assurance observations. It makes no claim to dominate these approaches and does not transport labels to unobserved deployment populations without additional assumptions.

Safe operating regions. Dixit et al. (2025) couple conformal perception calibration to a safe planner and evaluate navigation safety. Their guarantee concerns the resulting system, whereas our risk threshold alone says nothing about downstream control or clinical harm. Torens et al. (2025) monitor an operational design domain for an ML component; here uncertainty in average predictive loss determines an inner set within an explicitly supplied family. Hammar et al. (2026) learn a network's safe operating region through causal observation and intervention. Our present empirical analysis is offline and does not implement such active region expansion. These are meaningful precedents for region-based assurance, not evidence that the term SAR denotes a previously unstudied class of problems.

3 Target and Sampling Design

Let D_0 include all training and design information, including preprocessing estimates, learner randomization, a pilot sample if used, and the selected finite environment family $\mathcal{E}(D_0) = \{e_1, \ldots, e_K\}$. We condition on D_0 unless otherwise stated. The fitted pipeline g includes imputation, transformations, prediction, and any fixed decision threshold. For bounded loss $\ell \in [0, 1]$, define

$$R_g(e) = \mathbb{E}_{P_e}[\ell(Y, g(X_{\text{obs}}, M))], \qquad \mathcal{A}_\tau(g) = \{e \in \mathcal{E}(D_0) : R_g(e) \leq \tau\}. \tag{1}$$

The law P_e must be specified. Synthetic perturbation experiments establish results only for their artificial environment laws, not automatically for a real hospital or calendar period.

3.1 Paired stress experiments

Lemma 1 (Shared transformations) *Conditional on D_0, let (Z_i, V_i) be i.i.d. across i, independent of D_0, and let $h_k(\cdot; D_0)$ be fixed measurable maps into $[0, 1]$. Then $\mathbf{L}_i = (h_1(Z_i, V_i; D_0), \ldots, h_K(Z_i, V_i; D_0))$ are i.i.d. bounded random vectors. No independence across coordinates is required.*

Proof For any measurable vector sets $B_1, \ldots, B_n$, the events $\{\mathbf{L}_i \in B_i\}$ are measurable with respect to their respective independent (Z_i, V_i) rows, conditional on D_0. Their joint conditional probability factors. The same row law and fixed maps give the same vector distribution for each i. ■

Shared masks, noise, or outcome uniforms within a row are therefore compatible with simultaneous inference. Perfectly correlated coordinates are permitted. In contrast, normalizing missingness probabilities by the assurance-batch mean makes the transformation depend on other rows and is not justified by Lemma 1. Our corrected generators use fixed population-calibrated intercepts in the simulation. Repeated patients, temporal dependence, or cluster sampling also require an appropriate analysis rather than this i.i.d. argument. Conditioning on training data does not remove dependence among assurance rows.

3.2 Natural groups and independent samples

For a master sample of independent subjects, let G_{id} indicate a prespecified domain or subgroup. With $n_d = \sum_i G_{id}$ and $\widehat{p}_d = n_d/n$, estimate

$$\widehat{R}_{d,m} = \frac{\sum_i G_{id} L_i^{(m)}}{n_d}, \qquad \widehat{\psi}_{i,d,m} = \frac{G_{id}}{\widehat{p}_d}(L_i^{(m)} - \widehat{R}_{d,m}). \tag{2}$$

Common row multipliers applied to these influence estimates preserve overlapping-group and within-subject stress dependence. Fixed-dimensional asymptotic ratio inference additionally requires positive group probabilities and nondegenerate influence variances. Small or empty groups remain in the report and cannot certify under the prespecified minimum-size rule. For exact binomial inference, condition on a group's membership indicators: under i.i.d. master rows and a fixed group definition, its selected binary losses have the group-conditional Bernoulli law. A union bound can then cover all groups and stress conditions, despite overlap.

When environments instead have genuinely independent, unequal samples, the workflow accepts separate loss vectors. It neither truncates all samples to their shortest length nor pairs unrelated patients by row position. Validity remains a statement about the represented environment populations, not unobserved sites.

4 Risk Bounds and Set Containment

4.1 Finite-sample routes

Let $\widehat{R}_k$ be the mean of n_k independent bounded losses with expectation R_k. For binary losses, write $S_k = n_k \widehat{R}_k$. The one-sided exact binomial bound is

$$U_k^{\mathrm{CP}} = \begin{cases} Q_{1-\alpha/K}\{\mathrm{Beta}(S_k + 1, n_k - S_k)\}, & S_k < n_k, \\ 1, & S_k = n_k, \end{cases} \tag{3}$$

where Q denotes a quantile (Clopper and Pearson, 1934). For general losses in $[0, 1]$, use

$$U_k^{\mathrm{H}} = \min\left\{1, \widehat{R}_k + \sqrt{\frac{\log(K/\alpha)}{2n_k}}\right\}. \tag{4}$$

Proposition 2 (Finite-sample simultaneous assurance) *Conditional on the frozen design, either* (3)*, for Bernoulli samples, or* (4)*, for independent bounded samples, satisfies*

$$\mathbb{P}\{R_k \le U_k \text{ for every } k \mid D_0\} \ge 1 - \alpha. \tag{5}$$

Consequently, on this same event, $\widehat{\mathcal{A}}_\tau \subseteq \mathcal{A}_\tau$ *for every* $\tau \in [0,1]$ *simultaneously. Dependence between environment samples is allowed.*

Proof An exact one-sided binomial limit fails with probability at most α/K in each coordinate. For (4), the bounded-sum inequality of Hoeffding (1963) gives the same marginal failure bound. The union bound yields (5) without requiring independence between coordinates. On that event, $U_k \le \tau$ implies $R_k \le U_k \le \tau$ for every coordinate and every threshold. ■

These are standard constructions, stated here to make the software guarantee explicit. For example, $S_k = 0$ gives $U_k^{\mathrm{CP}} = 1 - (\alpha/K)^{1/n_k}$, a positive uncertainty bound despite no observed errors. Exact Holm uses $p_k = \mathbb{P}\{\mathrm{Bin}(n_k, \tau) \le S_k\}$ and the step-down rule of Holm (1979). These are valid lower-tail tests of $H_{0k} : R_k \ge \tau$ and yield finite-sample familywise containment at that fixed tolerance. The method is an environment-indexed LTT instantiation; it need not supply a full upper band.

4.2 Dependence-aware multiplier route

For paired rows, define $\widehat{\sigma}_k^2 = n^{-1}\sum_i (L_{ik} - \widehat{R}_k)^2$ and

$$T^* = \max_k \left\{ -\frac{1}{\sqrt{n}\widehat{\sigma}_k} \sum_{i=1}^{n} \xi_i (L_{ik} - \widehat{R}_k) \right\}, \quad \xi_i \overset{\text{iid}}{\sim} N(0,1). \tag{6}$$

Let $c^*_{1-\alpha}$ be its conditional quantile. The upper band is

$$U_k^{\mathrm{M}} = \min\{1, \widehat{R}_k + c^*_{1-\alpha}\widehat{\sigma}_k/\sqrt{n}\}. \tag{7}$$

Software enlarges negative simulated critical values to zero and assigns bound 1 to empirically constant columns. This avoids invalid studentization; it is not a general repair for rare-event approximation error.

Proposition 3 (Fixed-family asymptotic justification) *Under Lemma 1, for fixed* K *and positive population coordinate variances, the ideal conditional quantile in* (6) *yields* $\liminf_{n\to\infty} \mathbb{P}(R_k \le U_k^{\mathrm{M}} \text{ for all } k \mid D_0) \ge 1 - \alpha$*. For an empirical bootstrap quantile, the statement also requires the number of draws to increase sufficiently for quantile estimation error to vanish.*

Proof The multivariate central limit theorem applies to the independent bounded loss vectors. The sample covariance converges entrywise to the population covariance and each sample standard deviation converges to a positive constant. Conditional on the data, the vector in (6) before taking its maximum is exactly Gaussian with the empirical standardized covariance. Its distribution therefore converges to the same Gaussian limit as the standardized mean errors. The maximum of finitely many Gaussian coordinates with

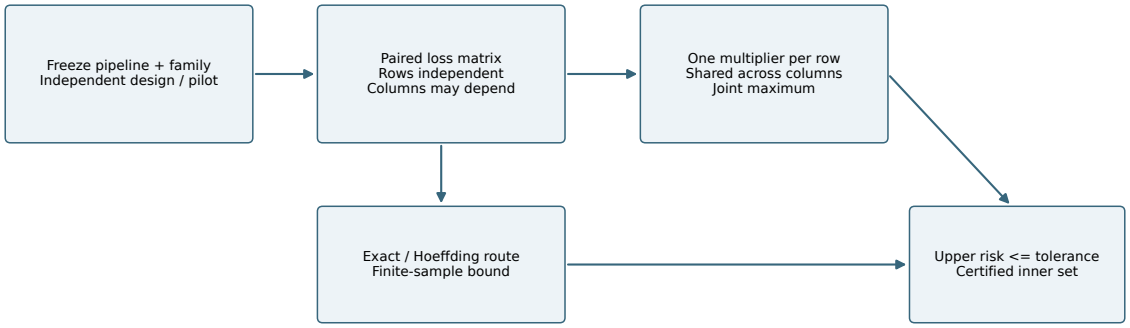


Figure 1: The workflow separates the sampling design from the choice of inferential bound. The multiplier route shares one random weight across all losses in a row. Exact and bounded-loss alternatives reach the same set-valued output with different guarantees.

nonzero marginal variance has a continuous distribution, even if the covariance is singular. Quantile consistency and the continuous mapping argument give asymptotic coverage. The set-containment implication then follows as in Proposition 2. ■

For increasing K, high-dimensional Gaussian approximation requires additional uniform moment and variance conditions; Appendix C records one conservative sufficient regime and later refinements. We do not use such asymptotic rates as practical sample-size prescriptions. The applications here with $K = 25$–125, including the 36–100-cell grid studies and the 72-condition MIMIC family, are treated as prespecified finite families; Proposition 3's fixed-family asymptotic statement is the relevant theoretical reference. Finite-sample validation remains decisive, especially for sparse binary errors.

Rare-event guidance. For Bernoulli loss, designs expected to yield fewer than roughly 5–10 errors in any environment should default to exact Clopper–Pearson Bonferroni or exact Holm certification rather than use the studentized Gaussian multiplier as the sole certificate. In the adverse binary design, risks ranged from 0.002 to 0.08 at $n = 500$, corresponding to roughly 1–40 expected errors per cell; multiplier coverage fell to 0.780, with the low-count part of this range motivating the 5–10-error diagnostic. The rule is practical guidance, not a theorem cutoff, and it is not sufficient by itself: the primary calibration also showed modest multiplier undercoverage, 0.928 with 95% Monte Carlo interval $[0.902, 0.948]$, even without a rare-cell trigger. Accordingly, exact Clopper–Pearson Bonferroni or exact Holm is our default for binary-loss certification; the multiplier may be reported as a dependence-aware secondary analysis when its approximation is supported by validation.

4.3 Conditioning, tuning, and repeated use

If a finite-sample statement holds conditional on every admissible training/design realization, averaging over D_0 preserves its error bound. Nested refits illustrate variation across

fitted pipelines; they do not transform an asymptotic method into a finite-sample theorem. Proposition 2 also clarifies a useful distinction: a fixed risk band supports all tolerances simultaneously, including a threshold inspected afterward. That mathematical fact does not justify retroactively defining an operationally meaningful risk requirement. Changing models, losses, families, or choosing the most favorable unadjusted method after assurance is different and requires protection over the enlarged selection problem.

Repeated 95% checks are not a time-uniform 95% statement. A simple conservative extension uses fresh assurance data at round t, conditions on the past, and applies a finite-sample procedure at $\alpha_t = \alpha/[t(t+1)]$. Since $\sum_{t\geq 1} \alpha_t = \alpha$, a union bound controls any erroneous certification over all rounds. This standard spending argument does not supply efficient adaptive or optimal sequential region learning.

5 Environment Design and Continuous Regions

Figure 2 provides a visual overview of the proposed statistical assurance framework.

The environment family is a scientific input. Begin with a specified loss and decision threshold, define interpretable axes and externally motivated ranges, and distinguish changes to the population from changes to measurement or observation. A transformation of one labeled sample is not evidence that an intended real deployment law is represented. Omitted axes remain outside the scope of the report.

Our independent-pilot procedure starts with a candidate grid, retains points near a pilot-estimated boundary together with coordinate extremes, and freezes that selected finite family before examining an independent assurance sample. Conditional on the pilot, the exact binomial construction applies without selective-inference machinery. Alternatively, one may first construct a valid band on an entire fixed superset and report a subset of it. Reusing the same outcomes to invent new environments and applying an unadjusted band only afterward is not justified by either argument.

For planning, (4) has additive width at most ϵ when $n \geq \log(K/\alpha)/(2\epsilon^2)$. This bound is conservative and controls band width only; it does not guarantee a particular assurance-region recall or a particular probability of certifying a cell. Pilot simulations can estimate recall for plausible risk margins, but they cannot establish external validity.

Higher-dimensional stress spaces. If d stress axes each use G grid levels, an exhaustive Cartesian design has $K = G^d$ environments. Thus even moderate resolution becomes expensive in data, computation, and interpretation as d grows. We recommend scientifically restricting axes when possible or using an independent pilot to screen a larger candidate family before freezing the assurance grid. Any pilot-selected family must remain independent of the assurance outcomes, or inference must protect a fixed superset. The current work does not provide an optimal active-grid search.

Suppose a compact continuous family has a scientifically justified global bound $|R(e) - R(e')| \leq L_R d(e, e')$. Define

$$U^{\mathrm{cont}}(e) = \min\left\{1, \min_{k\leq K}[U_k + L_R d(e, e_k)]\right\}. \tag{8}$$

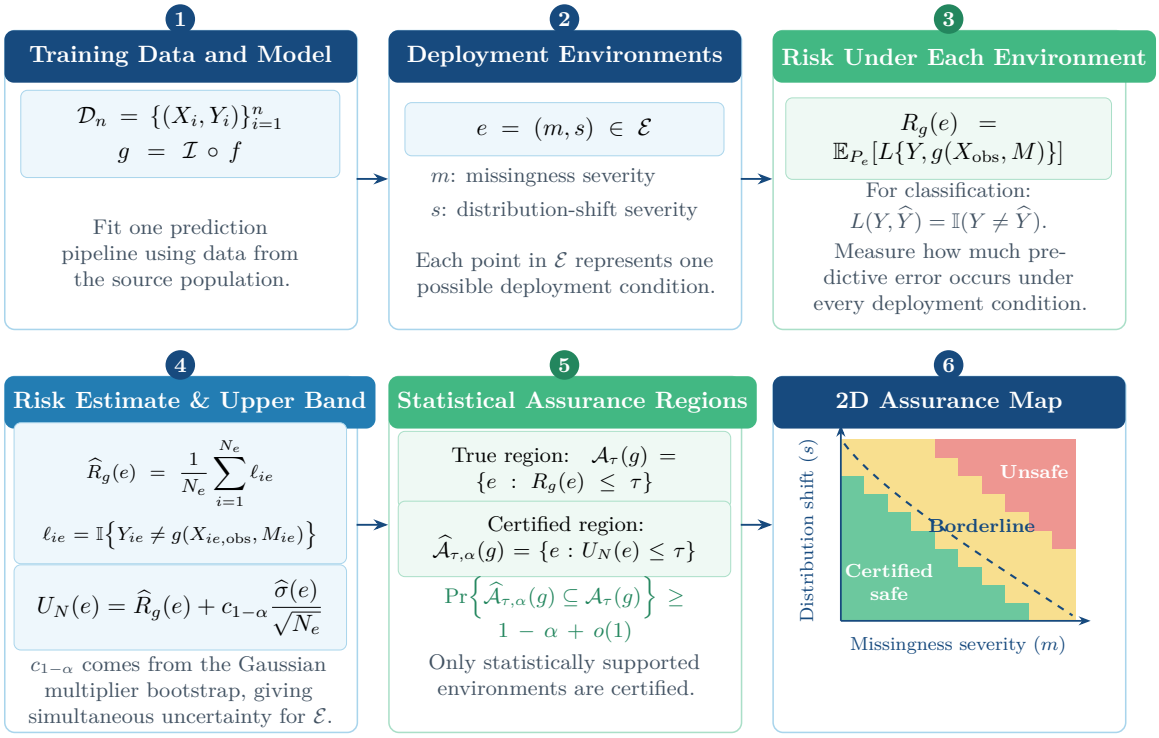


Figure 2: SciGuard statistical assurance framework. A predictive pipeline is evaluated over deployment environments indexed by missingness severity m and distribution-shift severity s. Environment-specific risks and simultaneous upper confidence bounds are used to construct a conservative certified assurance region. The resulting map is schematic rather than an empirical result and identifies deployment conditions under which the model is statistically supported to satisfy the prespecified risk threshold τ.

On the grid-coverage event, $R(e) \leq R(e_k) + L_R d(e, e_k) \leq U_k + L_R d(e, e_k)$ for every k, so (8) covers the risk everywhere. Thresholding it gives a continuous inner region. This is a standard smoothness consequence of simultaneous inference.

The largest observed slope on the assurance grid is not a valid global upper bound on L_R, and estimating L_R from the same outcomes used for certification invalidates the containment argument. When a data-driven value is unavoidable, our recommended workflow is sample splitting: use an independent pilot sample to propose and freeze a conservative smoothness bound, then use a separate assurance sample for the risk band. This removes same-sample selection bias but still cannot rule out unobserved rapid variation between design points without structural assumptions. If an externally or independently justified random $\widehat{L}_R$ bounds the global constant with failure probability δ, the combined error budget is at most $\alpha + \delta$. Without such a bound, finite-grid certification and explicitly assumption-dependent smoothness sensitivity are the defensible outputs.

6 Experimental Design and Evaluation

All reported results are generated by the shared implementation, with seed 20260927. They supersede the historical CSVs retained under `archive/pre_revision/`. Models, designs, and inferential settings are fixed in code; outcomes are not filtered to match earlier published-looking numbers. The full profile and smoke profile write separate directories. Tables and numerical macros are exported from full-run CSVs with a hash-based provenance record.

With known risk vector R, containment is $C = \mathbb{I}\{\widehat{\mathcal{A}}_\tau \subseteq \mathcal{A}_\tau\}$, ARR is $|\widehat{\mathcal{A}}_\tau \cap \mathcal{A}_\tau|/|\mathcal{A}_\tau|$, and UCP is $|\widehat{\mathcal{A}}_\tau \setminus \mathcal{A}_\tau|/|\widehat{\mathcal{A}}_\tau|$. We set UCP to zero for an empty certified set and leave ARR undefined when the true safe set is empty. Counts accompany these metrics. A high containment frequency can be vacuous when almost nothing is certified, and it does not imply that a full risk band covers at its advertised confidence level. Wilson intervals describe Monte Carlo uncertainty in binary coverage/containment indicators; mean recalls and paired differences have replication-level Monte Carlo standard errors (MCSEs).

6.1 Nested fitted-pipeline study

The central predictive-pipeline experiment uses 200 independent refits. Each generates 1,200 training rows, 400 source-calibration rows for ATC, 1,000 assurance rows, and 20000 separate reference rows. Ten Gaussian covariates have AR(1) covariance with parameter .35. Outcomes follow

$$\mathbb{P}(Y = 1 \mid X) = \operatorname{expit}\left[1.5\{.75X_0 - .65X_1 + .45X_2 + .55X_0X_1 - .35X_2^2 + .20X_3\}\right]. \quad (9)$$

A fixed normalized direction $d \propto (.8, -.6, .4, .2, 0, \ldots, 0)$ induces mean shift $s\Sigma^{1/2}d$, using the Cholesky factor. The grid has $m \in \{0, .1, .2, .3\}$ and $s \in \{0, .15, .30, .45\}$, with $\tau = .35$. Feature X_0 remains observed for comparability with the MAR study; severity m is the missingness rate among the other nine features. Imputation uses training means. The two pipelines are standardized logistic regression and histogram gradient boosting (60 boosting iterations, 15 maximum leaves, learning rate .08). The same base rows, label uniforms, and missingness randomness are used across models and environments.

The independent reference mean $\widetilde{R}_k$ is not called an oracle truth. Simultaneous two-sided exact binomial intervals $[a_k, b_k]$ allocate total error .01 over both models' reference risks within each refit. Define confirmed-safe cells by $b_k \le \tau$, confirmed-unsafe cells by $a_k > \tau$, and all others as ambiguous. Then

$$\begin{aligned} C^- &= \mathbb{I}\{\text{all certified cells are confirmed safe}\}, \\ C^+ &= \mathbb{I}\{\text{no certified cell is confirmed unsafe}\}. \end{aligned} \quad (10)$$

satisfy $C^- \le C \le C^+$ on that reference-coverage event. The means of these endpoints are reported as a reference bracket, not as confidence limits for a coverage probability. Point-reference containment and ARR remain labeled as such. The reference interval budget is per refit; it is not a joint .99 guarantee over all refits.

Table 1 gives point-reference containment 1.000 for all methods, which must not be read as proof of perfect containment. Logistic-regression cells near the boundary remain ambiguous under the reference intervals; the reference bracket is wider for the multiplier procedure. HGB has no ambiguous reference cells in these runs, making its favorable recall

Table 1: Nested refits. LR denotes logistic regression and HGB histogram gradient boosting. The reference bracket averages (10); it captures reference-risk ambiguity, not Monte Carlo uncertainty in a population coverage probability. ARR uses reference means.

Model	Method	Reference containment [95% CI]	ARR	Reference bracket
LR	Bonferroni normal	1.000 [0.981, 1.000]	0.167	[0.960, 1.000]
LR	Multiplier	1.000 [0.981, 1.000]	0.237	[0.940, 1.000]
LR	Bonferroni exact	1.000 [0.981, 1.000]	0.139	[0.970, 1.000]
LR	Hoeffding	1.000 [0.981, 1.000]	0.047	[0.995, 1.000]
LR	Exact Holm	1.000 [0.981, 1.000]	0.167	[0.955, 1.000]
HGB	Bonferroni normal	1.000 [0.981, 1.000]	0.669	[1.000, 1.000]
HGB	Multiplier	1.000 [0.981, 1.000]	0.735	[1.000, 1.000]
HGB	Bonferroni exact	1.000 [0.981, 1.000]	0.631	[1.000, 1.000]
HGB	Hoeffding	1.000 [0.981, 1.000]	0.395	[1.000, 1.000]
HGB	Exact Holm	1.000 [0.981, 1.000]	0.732	[1.000, 1.000]

Table 2: ATC-MC performance prediction, averaged over nested refits.

Model	MAE	Bias	RMSE
Logistic regression	0.031	-0.012	0.033
HistGradientBoosting	0.030	-0.014	0.032

a result for this particular fitted-model regime rather than evidence about difficult near-boundary clinical cases. The relative advantage over exact Holm is also model-dependent: the two methods have nearly equal HGB recall. Figure 3 makes these differences visible without assigning ATC a certification interpretation.

ATC uses maximum class probability and a source-calibrated strict threshold, with tied scores handled jointly. Table 2 reports its risk-prediction MAE, signed bias, and RMSE against the independent reference means. No target labels enter its risk predictions, whereas the certification methods use labeled assurance losses. This is a descriptive neighboring task, not a head-to-head comparison of equal guarantees.

6.2 Missingness-mechanism study

Across 100 logistic-regression refits, we compare MCAR, MAR, MNAR, and structured missingness with the same training and evaluation design. MAR depends only on always-observed X_0; MNAR depends on the latent value of the feature being hidden. Logistic intercepts solve a population Gaussian expectation equation for the target marginal missingness, using 64-node Gaussian quadrature. They are fixed before evaluation and are not normalized using assurance rows. Structured failure makes features 1–3 missing together, with every eligible feature retaining marginal severity m. The original cyclic driver scheme could hide its own conditioning variables and was not a valid general MAR implementation.

Reference intervals allocate error .01 over all four mechanisms within each refit. Table 3 shows that the reference uncertainty can matter even when every point-reference containment indicator equals one. In particular, ambiguous cells reduce the MCAR multiplier lower

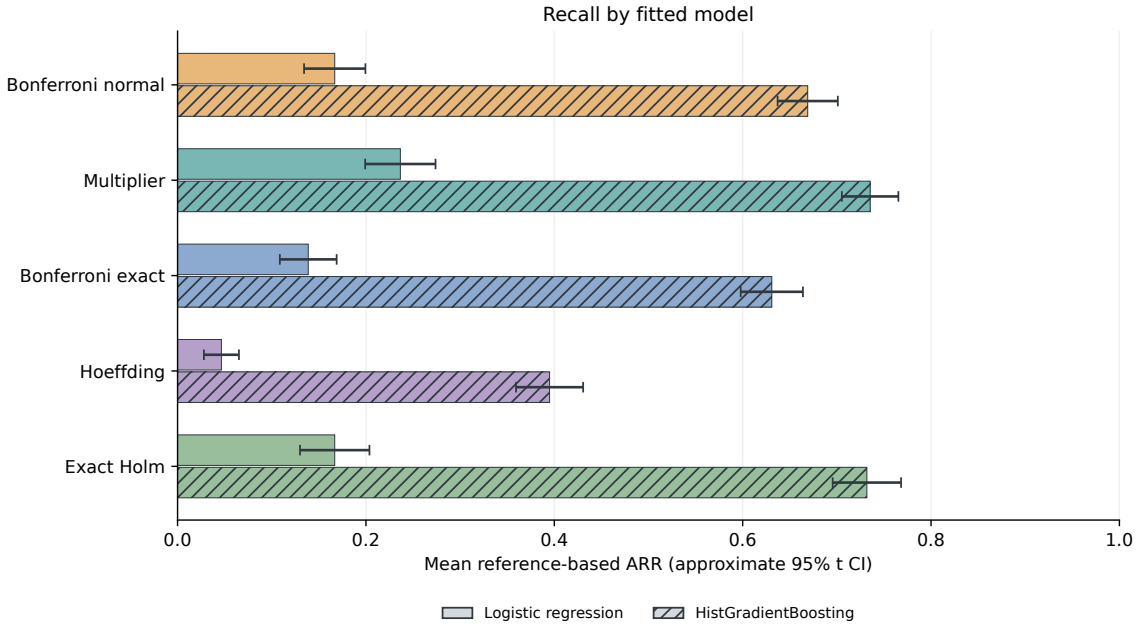


Figure 3: Reference-based recall across nested refits. Bars are means; error bars are approximate pointwise 95% t intervals across independent refits. Colors identify certification methods and hatching identifies the fitted model. These intervals quantify Monte Carlo uncertainty in mean recall, not uncertainty in the reference risks.

Table 3: Multiplier results for missingness mechanisms. Ambiguous cells are averaged over the 16 environments per refit. All metrics use the corrected mechanism definitions.

Mechanism	Reference containment	Reference bracket	ARR	Ambiguous cells
MCAR	1.000 [0.963, 1.000]	[0.940, 1.000]	0.211	1.83
MAR	1.000 [0.963, 1.000]	[0.990, 1.000]	0.249	0.28
MNAR	1.000 [0.963, 1.000]	[1.000, 1.000]	0.303	0.00
Structured	1.000 [0.963, 1.000]	[0.930, 1.000]	0.232	0.98

endpoint. These results concern the chosen parameterizations; they do not imply that MNAR is generally easier or safer than MCAR. Realized missingness fractions and all exact-baseline comparisons are retained in the current-run CSVs.

6.3 Calibration with exactly known risks

To isolate inference, a Gaussian copula generates binary losses with the 25 marginal risks stored in the supplied simulation surface, $n = 1000$, latent correlation .55, and $\tau = .34$. We run 500 replications and 2,000 multiplier draws per replication. This is a calibration

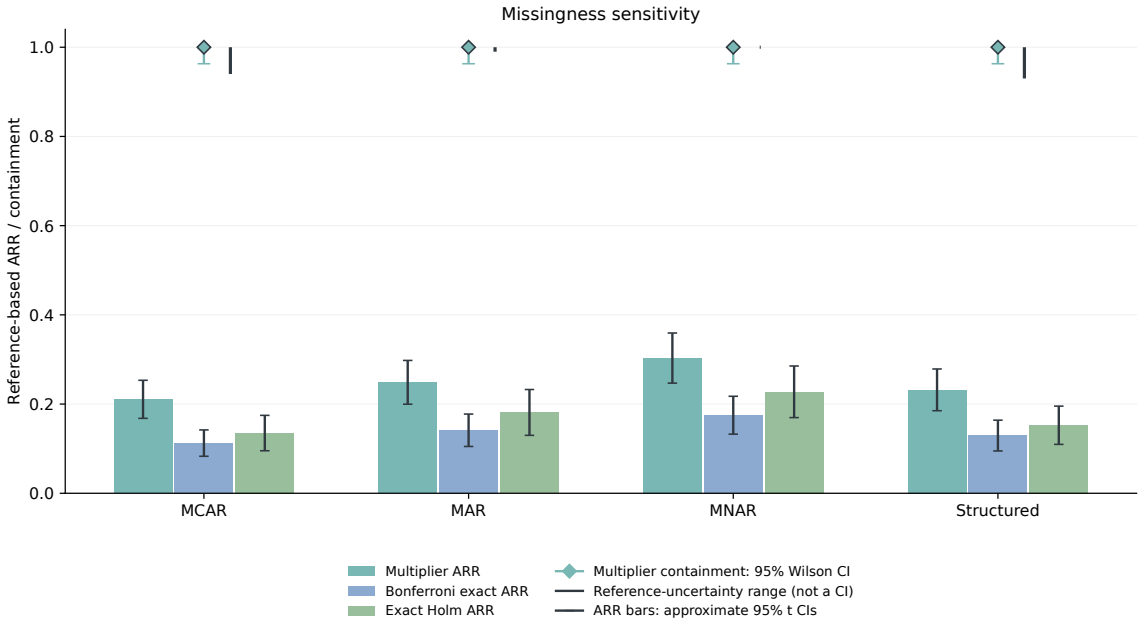


Figure 4: Mechanism sensitivity and the effect of reference-risk uncertainty. ARR bars have approximate pointwise 95% t intervals across refits. Multiplier point-reference containment has 95% Wilson intervals. Separate black vertical segments span mean lower/upper containment indicators from reference-risk uncertainty; these segments are not confidence intervals.

Table 4: Primary calibration study. Coverage is simultaneous risk-band coverage. Holm supplies fixed-tolerance tests rather than a risk band, so its coverage entry is inapplicable.

Method	Coverage [95% CI]	Containment [95% CI]	ARR	UCP
Pointwise normal	0.514 [0.470, 0.558]	0.958 [0.937, 0.972]	0.522	0.0042
Bonferroni normal	0.938 [0.913, 0.956]	0.998 [0.989, 1.000]	0.282	0.0002
Multiplier	0.928 [0.902, 0.948]	0.998 [0.989, 1.000]	0.297	0.0002
Bonferroni exact	0.960 [0.939, 0.974]	0.998 [0.989, 1.000]	0.258	0.0002
Hoeffding	1.000 [0.992, 1.000]	1.000 [0.992, 1.000]	0.138	0.0000
Exact Holm	–	0.998 [0.989, 1.000]	0.270	0.0002

experiment, not a fitted-pipeline deployment study. All methods receive the same loss matrix in each replication.

Pointwise normal intervals attain only 0.514 simultaneous coverage. The multiplier band attains 0.928, compared with 0.960 for exact Bonferroni (Table 4). The multiplier's mean ARR is 0.297, compared with 0.258 for exact Bonferroni. This efficiency difference accompanies different finite-sample validity: the multiplier is not a finite-sample replacement

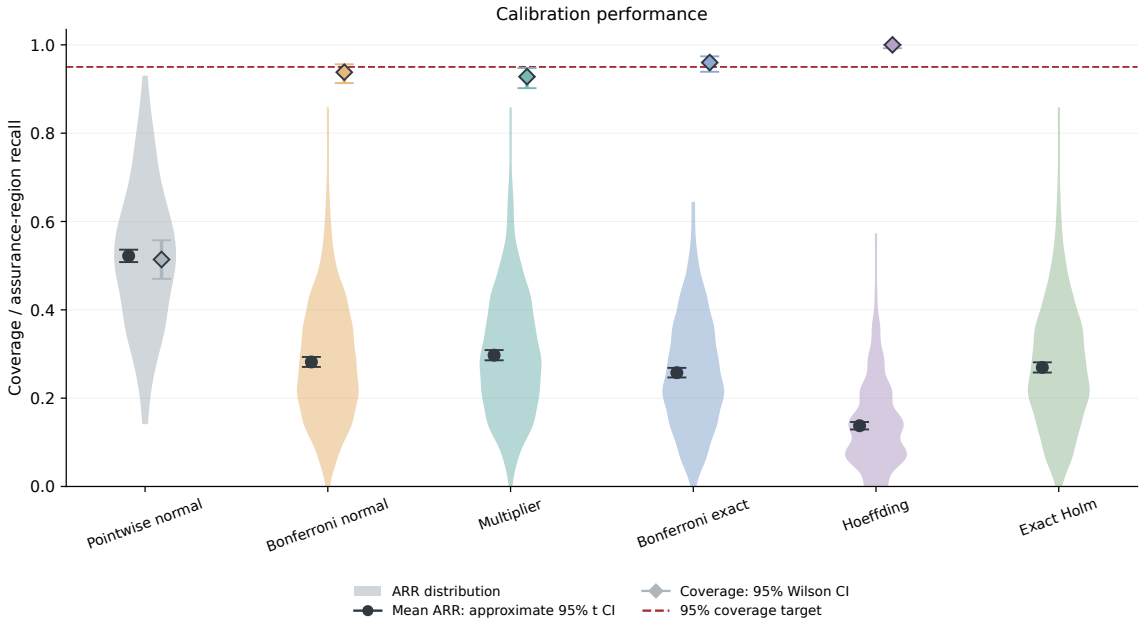


Figure 5: Simultaneous coverage and the recall distribution over primary replications. Coverage diamonds have pointwise 95% Wilson intervals; black markers show mean ARR with approximate pointwise 95% t intervals, overlaid on the ARR distributions. The dashed line is the nominal coverage target; higher recall alone does not establish validity.

for exact inference. Its primary coverage estimate lies below .95; the Monte Carlo interval describes the precision of that finding and does not verify a universal nominal guarantee.

6.4 Dependence, sample size, and adverse regimes

The sensitivity study uses 300 replications per setting, varying $n \in \{500, 1000, 5000\}$ at latent correlation .55 and correlation $\rho \in \{.10, .55, .90\}$ at $n = 1000$. Table 10 reports paired recall gains, not differences between independently simulated studies. Against normal Bonferroni the gain is negligible at low correlation and larger at high correlation. Comparison with exact Holm is necessary: exploiting dependence is not the only way to reduce multiplicity conservatism.

Three additional 200-replication settings challenge the main result: independent Beta losses with concentration four and known means from the main surface; binary risks equally spaced from .002 to .08 at $n = 500$, $K = 25$, and $\tau = .035$; and a $K = 125$ binary family at $n = 1000$ formed by repeating the main risk values with new copula-specific noise. The last setting enlarges the number of hypotheses, not the dimensionality of continuous deployment geometry. Losses remain bounded; no robustness claim is made for unbounded heavy tails or adversarially selected environments.

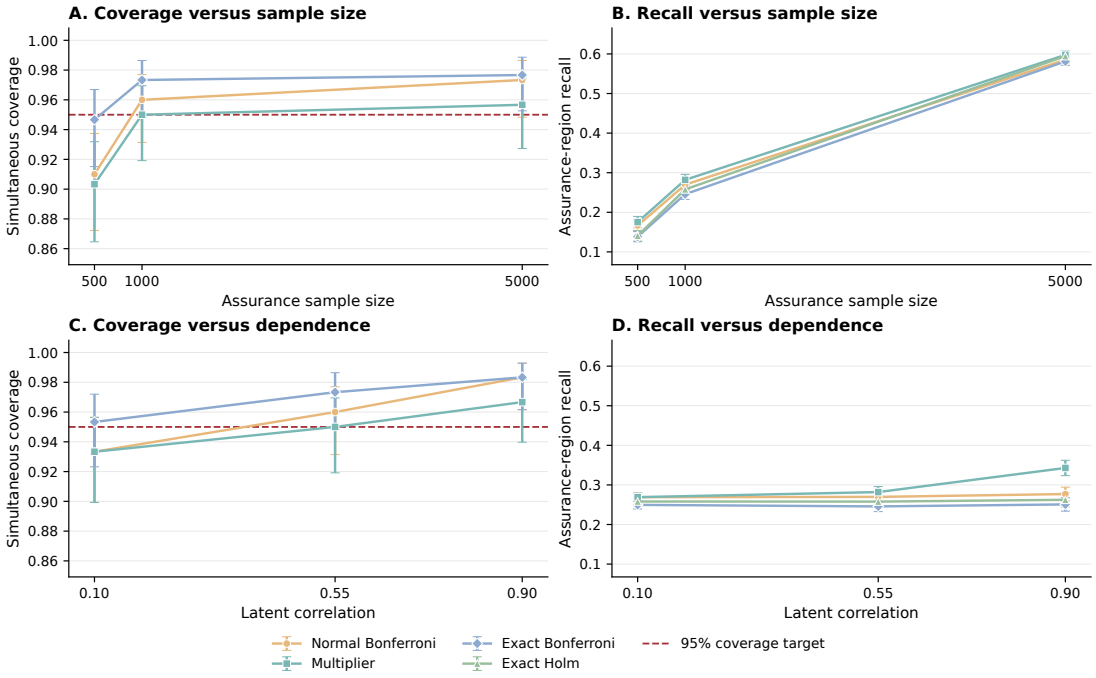


Figure 6: Coverage and recall sensitivity. Latent Gaussian correlation is a generator parameter, not the observed binary-loss correlation. Coverage error bars are pointwise 95% Wilson intervals; recall error bars are mean ±1.96 MCSE. Vertical scales are shared within each metric and include interval endpoints. Holm is omitted from risk-band coverage panels.

The rare-event multiplier coverage is 0.780 (Table 5). Refusing to certify zero-variance columns does not prevent this failure in low-count, nonconstant columns. Exact binomial methods remain applicable, and Hoeffding can be very conservative. These adverse results provide a practical reason to prefer a finite-sample route when such control is required. Selecting whichever unadjusted bound yields the most certificates after seeing assurance losses is not a valid remedy.

6.5 Continuous example and independent pilot

The illustrative surface $R(m, s) = .24 + .12m + .10s + .05ms$ on $[0, .4] \times [0, .6]$ has a valid Euclidean Lipschitz bound $L_R = \sqrt{.15^2 + .12^2}$. We use a 5×5 assurance grid, $\tau = .32$, 200 replications, and an 81×81 mesh to compute numerical recall and containment diagnostics. Table 6 distinguishes grid coverage from mesh containment. The analytical argument for (8), not the dense mesh, supplies coverage over every point when the grid band and the global bound are valid. The 81×81 mesh recall and containment values are numerical diagnostics only: they approximate geometric behavior and do not provide or replace an inferential guarantee. Separate exports evaluate sensitivity to half and twice the analytic bound; the half-bound is not asserted valid.

Table 5: Adverse-regime results. Exact binomial bounds are not applicable to continuous losses.

Scenario	Method	Coverage [95% CI]	Containment	ARR
Continuous bounded	Bonferroni normal	0.925 [0.880, 0.954]	1.000	0.469
Continuous bounded	Multiplier	0.925 [0.880, 0.954]	1.000	0.470
Continuous bounded	Hoeffding	1.000 [0.981, 1.000]	1.000	0.008
Rare binary	Bonferroni normal	0.780 [0.718, 0.832]	0.995	0.455
Rare binary	Multiplier	0.780 [0.718, 0.832]	0.990	0.502
Rare binary	Bonferroni exact	0.995 [0.972, 0.999]	1.000	0.379
Rare binary	Hoeffding	1.000 [0.981, 1.000]	1.000	0.000
Larger binary family	Bonferroni normal	0.945 [0.904, 0.969]	0.995	0.208
Larger binary family	Multiplier	0.910 [0.862, 0.942]	0.990	0.232
Larger binary family	Bonferroni exact	0.960 [0.923, 0.980]	0.995	0.188
Larger binary family	Hoeffding	0.995 [0.972, 0.999]	1.000	0.088

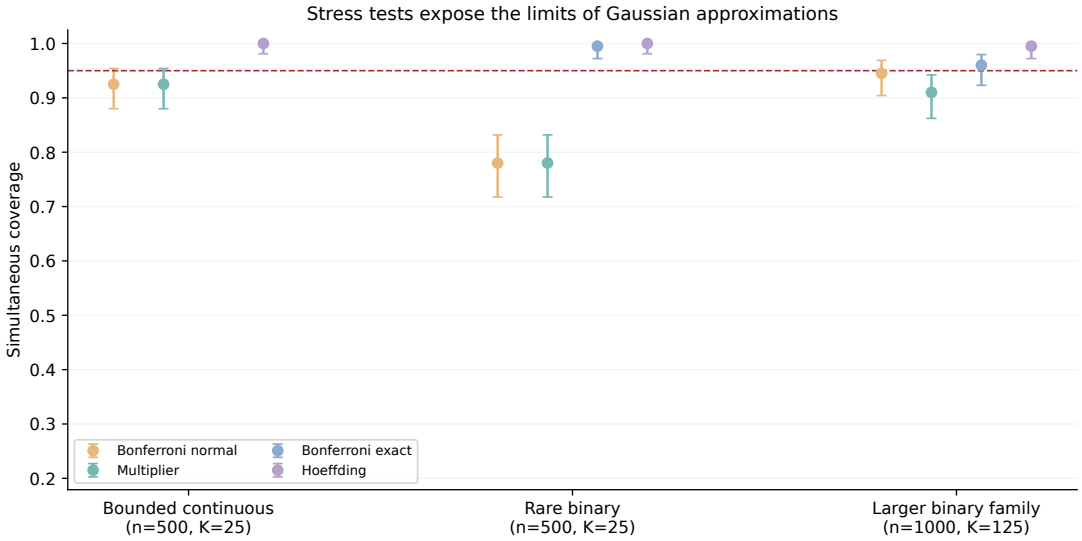


Figure 7: Coverage failures in adverse regimes, with 95% Wilson intervals.

The pilot example starts with an 11×11 candidate grid on the same rectangle. Using 300 independent pilot rows, it retains points whose pilot risks are within .015 of .32 and coordinate extrema, then certifies the selected family on new assurance rows with exact Bonferroni. Two hundred replications and a sample-size planning table are exported. This demonstrates a valid split-sample workflow; it does not establish an optimal adaptive grid or certification of all unselected points.

7 Wisconsin Diagnostic Breast Cancer Stress Test

The supplied WDBC data contain 569 observations. A fixed stratified 65/35 split produces 369 training rows and 200 assurance rows. The frozen models are standardized logistic

Table 6: Continuous example. Mesh recall is an equal-grid-point area approximation.

Method	Grid coverage	Mesh containment	Mesh recall	Mesh UCP
Multiplier	0.940	0.975	0.460	0.0004
Bonferroni exact	0.965	0.980	0.404	0.0001
Hoeffding	1.000	1.000	0.185	0.0000

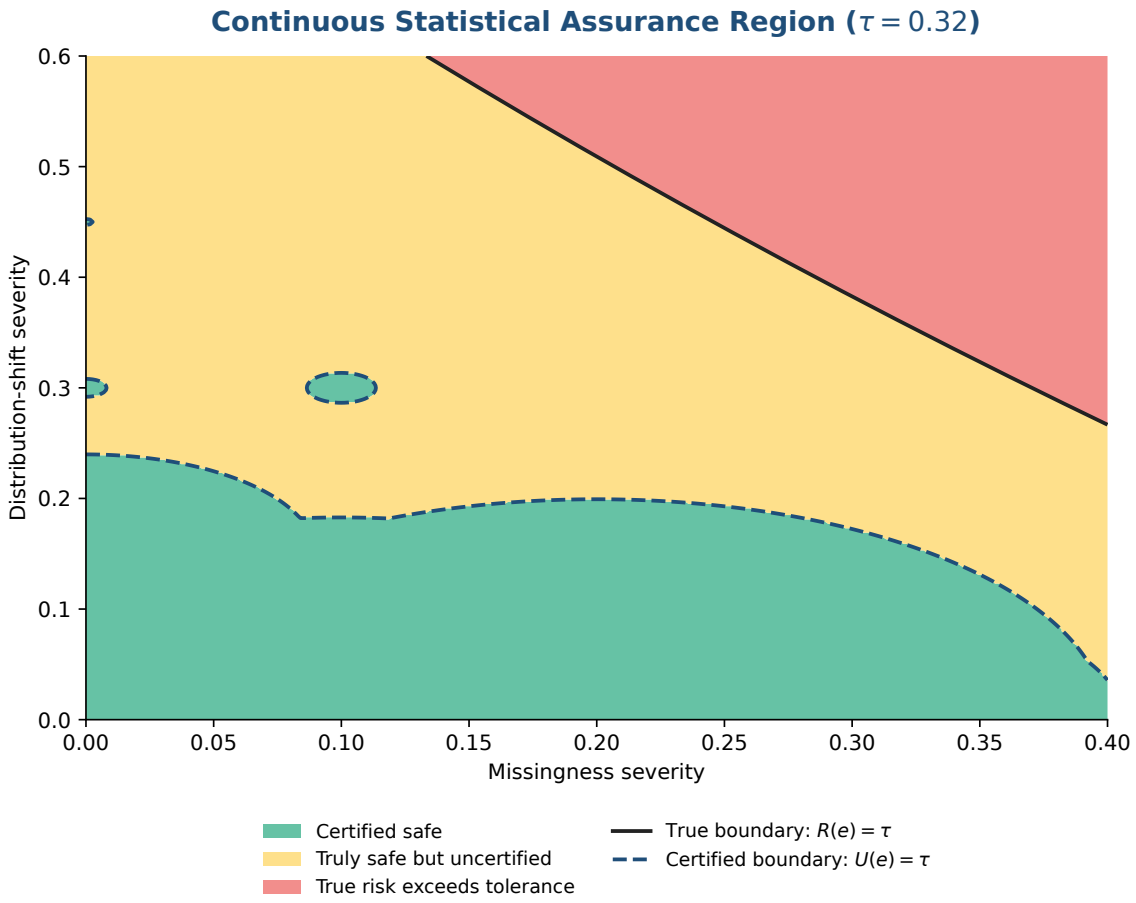


Figure 8: One fixed multiplier replication of the continuous SAR. Green denotes certified points within the analytic safe region; yellow denotes analytically safe but uncertified points, and salmon denotes points above the risk tolerance $\tau = .32$. The solid curve is the analytic boundary $R(e) = \tau$ and the dashed curve is the certified boundary $U(e) = \tau$. The Lipschitz envelope is evaluated directly on a dense mesh; contour interpolation is for display only. Multiplier validity is asymptotic.

regression, an Extra Trees ensemble with 500 trees, and histogram gradient boosting. Training means impute missing values; Gaussian feature noise is scaled by training standard deviations. One noise matrix and one uniform missingness matrix are shared across stress

Table 7: WDBC results for separate model-specific 36-environment families. The nominal observed risk is descriptive. Counts at two tolerances do not establish clinical safety.

Model	Method	Nominal risk	Certified $\tau = .08$	Certified $\tau = .10$
LR	Multiplier	0.010	23/36	30/36
LR	Bonferroni normal	0.010	20/36	29/36
LR	Bonferroni exact	0.010	15/36	23/36
LR	Exact Holm	0.010	15/36	27/36
Extra Trees	Multiplier	0.035	9/36	19/36
Extra Trees	Bonferroni normal	0.035	2/36	17/36
Extra Trees	Bonferroni exact	0.035	0/36	9/36
Extra Trees	Exact Holm	0.035	0/36	9/36
HGB	Multiplier	0.045	2/36	14/36
HGB	Bonferroni normal	0.045	0/36	12/36
HGB	Bonferroni exact	0.045	0/36	2/36
HGB	Exact Holm	0.045	0/36	2/36

conditions and models. The primary family uses $m \in \{0, .05, .10, .20, .30, .40\}$ and $s \in \{0, .10, .20, .30, .40, .50\}$. The multiplier uses 5,000 draws. Table 7 includes exact procedures so that Gaussian and finite-sample answers are not conflated in this small sample.

Logistic regression has the lowest nominal observed error in this split, but counts depend strongly on the method and tolerance. These are separate model-specific simultaneous statements: choosing a model by comparing these bands does not confer joint model-selection coverage. The tolerances are illustrative, not medical standards. WDBC perturbations are synthetic and do not represent observed temporal or inter-hospital shifts; its single split cannot estimate the true deployment safe set, ARR, or a coverage probability.

Equally spaced 4×4, 6×6, and 10×10 logistic grids span the same rectangle. Their results appear in Table 11. The primary 36-cell grid is nonuniform in missingness and is not identical to the equally spaced 6×6 grid. Certified grid fractions cannot be interpreted as continuous areas without an additional design measure or smoothness argument. A grid is not selected retrospectively for the largest certificate.

8 MIMIC-IV External Benchmark

We executed the local protocol on credentialed MIMIC-IV v3.1 (Johnson et al., 2024, 2023). The purpose of this experiment is to test whether the assurance workflow can be carried through on a large real EHR cohort with approximate temporal grouping and native missingness. We do not establish or quantify covariate shift or outcome shift across these groups, and the experiment is not a claim that the fitted mortality models or either risk target constitute a clinically validated deployment rule. The primary 0–1-loss specification is intentionally permissive and certifies every adequately sized condition, including the trivial always-survive comparator. A secondary class-sensitive specification, fixed before computing its deployment certificates, instead targets false-negative risk and yields a selective assurance

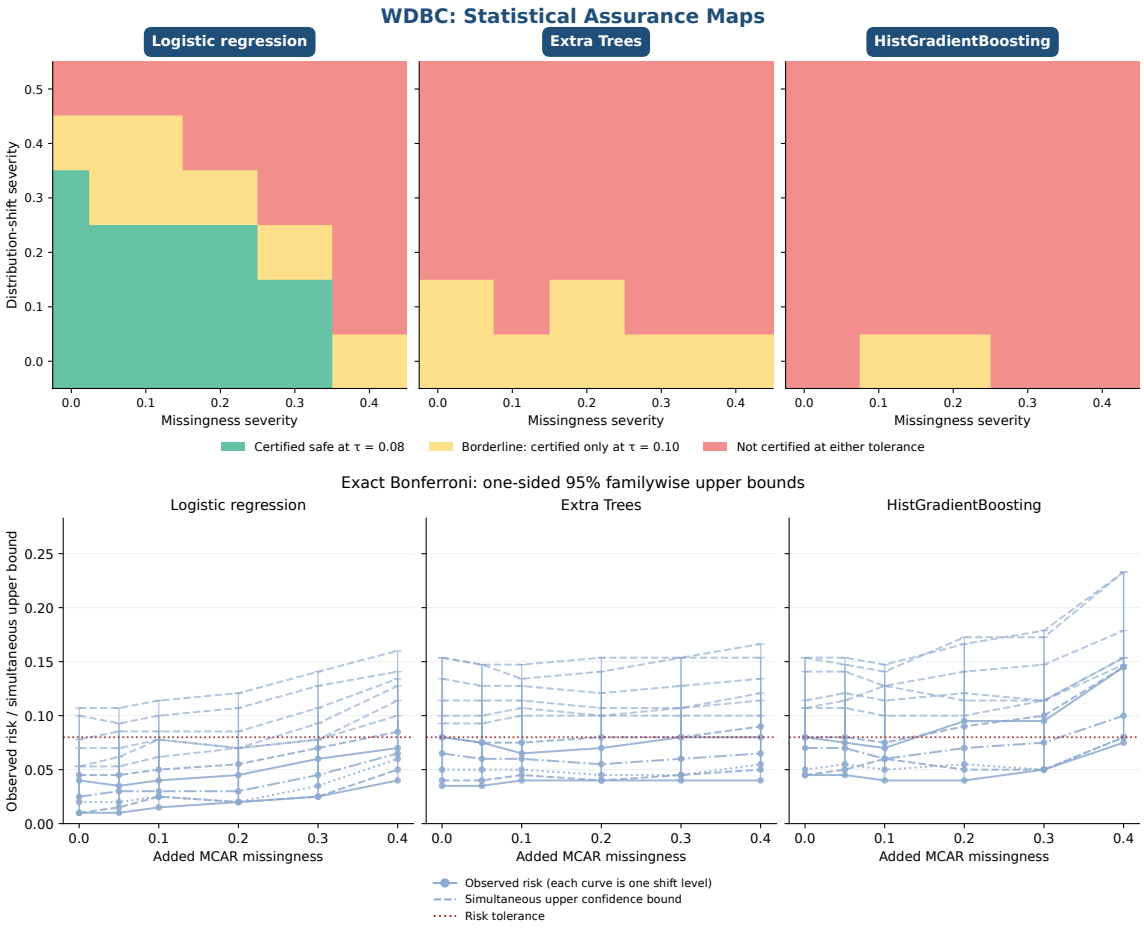


Figure 9: WDBC assurance maps (top) and exact Bonferroni upper-risk plots (bottom) for three models. Green cells certify at $\tau = .08$; yellow cells certify only at $\tau = .10$; salmon cells certify at neither tolerance. Non-certification does not establish unsafe true risk. In the lower plots, markers show observed risk and dashed curves show one-sided 95% familywise upper confidence bounds; the dotted line is $\tau = .08$. Each curve corresponds to one shift level.

region. Together the two specifications demonstrate both execution of the workflow and the consequences of target design.

Cohort and prediction task. The raw join contained 546,028 admissions. After retaining valid adult admissions and selecting the first eligible admission per patient, 223,452 patients remained. Admission-aligned temporal intervals placed 106,864 patients in the source cohort and 35,114 in deployment; 81,474 admissions with source/deployment ambiguity under the padded interval rule were excluded. The ambiguity exclusion is a deterministic row-level function of each patient's own shifted interval metadata and fixed cutoffs; it does not couple different patients, so after selecting one admission per patient the retained observations

preserve the independent-row construction used in Lemma 1. The source cohort was split into 85,491 fitting observations and a 21,373-observation source holdout. The binary target was in-hospital mortality. Predictors were age, an age-topcoding indicator, gender, admission type, admission location, insurance, marital status, and race. These variables are intentionally restricted to demographic and administrative fields available from the admission-level tables; no laboratory measurements, vital signs, medications, or ICU time-series variables enter the models. Identifiers, outcome flags, death/discharge times, and length of stay were not used as predictors.

MIMIC-IV shifts dates separately for each patient. We therefore align each admission to the patient anchor year, form approximate admission intervals, pad interval endpoints by one year for boundary ambiguity, and restrict them to the declared v3.1 coverage. Because the dates are deidentified at the patient level, these labels are approximate ordering intervals rather than literal shared calendar periods and should not be interpreted as exact secular trends. We do not use these labels as evidence that a particular magnitude of covariate or outcome shift occurred; they define prespecified deployment groups for the assurance exercise. The source period ends by 2016. Deployment groups are the resulting post-cutoff approximate intervals 2017–2021, 2018–2022, 2019–2022, 2020–2022, 2021–2022, and 2022–2022. The interval labels may overlap in calendar time, but each retained patient contributes to one prespecified deployment-domain label. The grouped construction in Section 3.2 accommodates unequal domain sizes, while common-row inference preserves dependence across the model and missingness conditions evaluated on the same patients.

Prespecified family and certification rule. We fit unweighted logistic regression and histogram gradient boosting on the source training cohort and retain an always-survive comparator. Added MCAR feature missingness levels were $m \in \{0, .10, .20, .30\}$, with native missingness recorded separately. The three models, six temporal domains, and four missingness levels define one 72-condition family. We fixed $\alpha = .05$, 5,000 multiplier draws, and a minimum group size of 200 before certification. The operational tolerance was fixed as $\tau = .35$ for 0–1 classification error before examining deployment outcomes. It was chosen deliberately as a permissive workflow-execution threshold, not from the deployment results and not as a clinically acceptable mortality-error rate. Because the source-holdout risks were already available when this operational value was finalized, the primary specification should not be interpreted as a prospective clinical requirement. Multiplier, exact-Bonferroni, Hoeffding, and Holm procedures were all applied to the same family.

Results. The source-holdout 0–1 risks were 0.0184 for logistic regression, 0.0185 for histogram gradient boosting, and 0.0184 for the always-survive comparator. Five temporal domains met the minimum group size, yielding 60 eligible model-domain-missingness conditions; the 2022–2022 group contained only 32 patients, so its 12 conditions were retained as ineligible and not certified. All 60 eligible conditions were certified by all four inferential procedures. For exact Bonferroni, the largest upper risk among eligible conditions was 0.0474, far below the prespecified tolerance; the largest Hoeffding upper bound was 0.1044. The multiplier critical value was 2.538 and its largest eligible upper bound was 0.0465. Figure 10 displays the exact-Bonferroni assurance map, and Table 8 summarizes the temporal groups.

Interpretation under class imbalance. The primary run exposes an important limitation of thresholded 0–1 loss for a rare outcome. At zero added missingness, logistic-regression

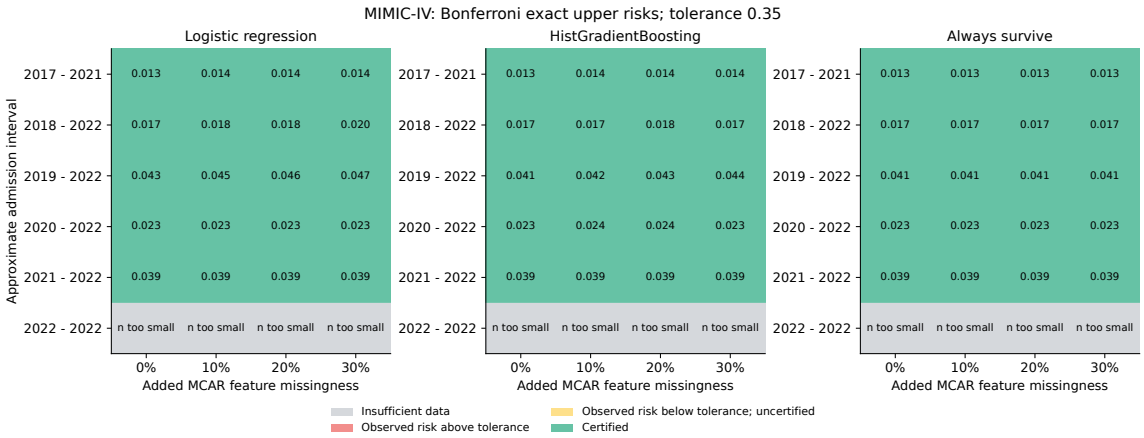


Figure 10: MIMIC-IV v3.1 assurance map using familywise exact-Bonferroni upper risks at $\tau = .35$. Columns add 0–30% MCAR feature missingness; rows are approximate admission intervals. Green cells are certified. The 2022–2022 interval has $n = 32 < 200$ and is withheld from certification. The figure reports statistical support for the prespecified 0–1-risk criterion, not clinical safety.

Table 8: MIMIC-IV temporal-domain summary. Mortality prevalence, logistic-regression risk, AUROC, and AUPRC are shown at zero added missingness. "Max exact upper" is the largest exact-Bonferroni upper risk across the three models and four missingness levels in that domain. The last column gives certified/total model-missingness conditions.

Interval	n	Mort.	LR risk	AUROC	AUPRC	Max U	Cert.
2017 - 2021	2,312	0.006	0.006	0.805	0.099	0.014	12/12
2018 - 2022	1,645	0.007	0.007	0.871	0.069	0.020	12/12
2019 - 2022	29,419	0.037	0.039	0.844	0.176	0.047	12/12
2020 - 2022	1,282	0.010	0.010	0.914	0.104	0.024	12/12
2021 - 2022	424	0.012	0.012	0.951	0.173	0.039	12/12
2022 - 2022	32	0.031	0.031	0.935	0.333	–	0/12[†]

[†]Ineligible because $n < 200$; retained but not certified.

AUROC ranged from 0.805 to 0.951 across the five eligible temporal domains, yet sensitivity at the default 0.5 prediction threshold was zero in four domains and only 0.049 in 2019–2022. The always-survive comparator was also certified in every eligible condition because mortality prevalence was low and $\tau = .35$ was permissive relative to the observed 0–1 risks. Thus the primary certificate is internally valid for the specified loss and tolerance while remaining weak as a statement about mortality detection.

Secondary class-sensitive specification. To make the real-data assurance target informative for the minority outcome, we report a secondary analysis based on false-negative risk,

$R_{\mathrm{FN}}(e) = \Pr\{\widehat{Y} = 0 \mid Y = 1, e\}$. Before computing these secondary deployment certificates, we fixed $\tau_{\mathrm{FN}} = .20$, corresponding to an operational requirement of sensitivity at least 0.80; this 80% requirement is interpretable but is not claimed to be a clinical standard. For each learned model, the largest classification threshold attaining at least 0.80 sensitivity was selected using only the independent source holdout and then frozen: 0.0217 for logistic regression and 0.0227 for histogram gradient boosting, each attaining source sensitivity 0.802 with specificity 0.747 and 0.749, respectively. Because the source holdout is disjoint from the deployment cohort, this threshold calibration is part of the frozen design information and does not reuse deployment assurance outcomes. The always-survive comparator has sensitivity zero by construction.

Because the effective sample size is now the number of deaths in each domain, exact Clopper–Pearson Bonferroni is the primary certification route and exact Holm is retained as a secondary check. The secondary protocol fixed a minimum of two positive cases per domain before the deployment certificates were computed. Positive-case counts in the six approximate domains were 14, 12, 1,092, 13, 5, and 1; the final 2022–2022 domain was therefore retained as ineligible for this specification. Among the 60 eligible model-domain-missingness conditions, only 3 were certified by exact Bonferroni, all for histogram gradient boosting in the 2019–2022 group at 0%, 10%, and 20% added missingness. Their observed sensitivities were 0.852, 0.848, and 0.845, with simultaneous upper false-negative-risk bounds 0.1855, 0.1895, and 0.1925. At 30% missingness the same model retained observed sensitivity 0.818, but its upper bound increased to 0.218 and the cell was not certified. Logistic regression certified 0 of 20 eligible conditions, and the always-survive comparator certified 0 of 20.

The 3-of-60 count should be interpreted primarily as a statement about where the data have enough positive-outcome information to support a familywise certificate, not as a ranking of model quality across temporal domains. Four of the five eligible domains contain only 5–14 deaths. With exact Bonferroni correction over the full 72-condition family (the 12 ineligible conditions remain in the multiplicity family), even zero observed false negatives among 14 positive cases gives a one-sided upper bound of approximately 0.405, well above $\tau_{\mathrm{FN}} = .20$; the smaller positive-case domains are still less precise. Thus no cell in those four domains could certify at this tolerance even under perfect observed sensitivity. All three certificates arise in the 2019–2022 domain, which contains 1,092 positive cases. Non-certification in the remaining small-positive domains is therefore power-limited and should not be interpreted as evidence that their fitted models are worse. Figure 11 and Table 9 summarize this secondary specification.

The contrast between the two MIMIC specifications is the substantive design lesson: deployment assurance cannot repair a scientifically inadequate loss, threshold, or tolerance, but changing the scientific target changes the assurance question in a transparent way. The two specifications are reported as separate assurance statements and no cross-specification multiplicity adjustment is claimed. The secondary specification was introduced after the primary workflow had been executed; neither specification is selected post hoc as the "better" certificate, and the two should not be combined into a single familywise statement. The secondary result is still an assurance analysis rather than evidence of clinical safety.

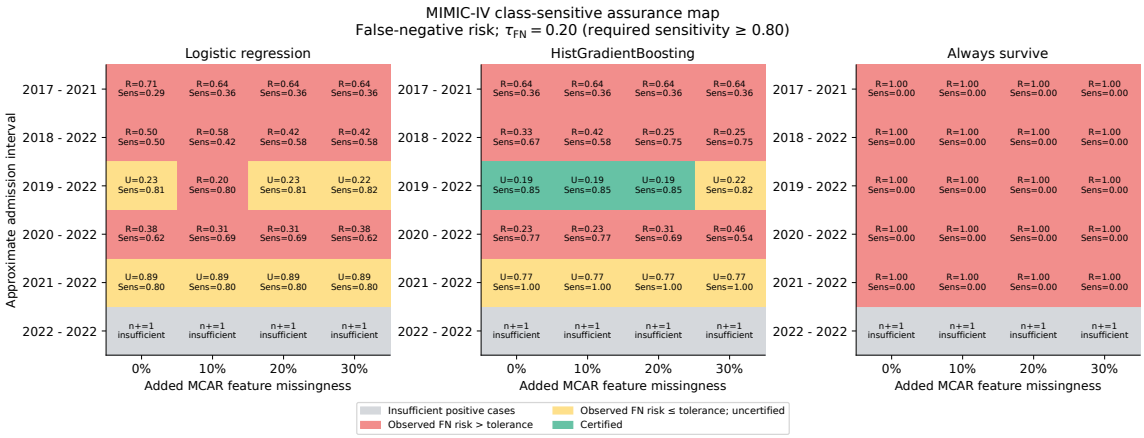


Figure 11: MIMIC-IV class-sensitive assurance map for false-negative risk at $\tau_{\mathrm{FN}} = .20$, equivalent to a required sensitivity of at least 0.80. Thresholds for the two learned models were calibrated on the independent source holdout and then frozen. Exact Clopper–Pearson Bonferroni is the primary certification route. Green cells are certified, yellow cells have observed false-negative risk at or below tolerance but an upper bound above tolerance, red cells have observed false-negative risk above tolerance, and gray cells have insufficient positive cases. Only three conditions are certified, all for histogram gradient boosting in the 2019–2022 group at 0–20% added missingness.

Table 9: Secondary class-sensitive MIMIC-IV specification. "Threshold" is the frozen classification threshold selected on the source holdout to attain sensitivity at least 0.80. Each learned model has 20 eligible deployment conditions (five domains by four missingness levels). Eligibility required at least two positive cases per domain, fixed in the secondary protocol before certification; the one-positive-case 2022–2022 domain is therefore excluded.

Model	Threshold	Source sens.	Source spec.	Certified/eligible
Logistic regression	0.0217	0.802	0.747	0/20
HistGradientBoosting	0.0227	0.802	0.749	3/20
Always survive	–	0.000	1.000	0/20

Only aggregate tables, the frozen configuration, and source-file hashes are distributed. Credentialed patient-level MIMIC-IV data, patient-level derivatives, and fitted clinical models are not bundled.

9 Practical Interpretation and Limitations

Choosing a method. For binary losses, we recommend exact Clopper–Pearson Bonferroni or exact Holm as the default certification route when its sampling assumptions are defensible.

Sparse counts strengthen that recommendation: fewer than roughly 5–10 expected errors in any cell should trigger an exact route automatically. The multiplier remains useful as a dependence-aware secondary analysis, but the primary calibration's 0.928 simultaneous coverage shows that modest undercoverage can occur even when sparsity is not the obvious failure mode. Use a bounded-loss inequality for nonbinary bounded losses when conservatism is acceptable. Any multiplier use should be accompanied by finite-sample validation in a design comparable to the intended application, and any method selection performed after seeing assurance losses must itself be covered by the multiplicity control.

Scalability. Grid complexity grows multiplicatively across deployment axes: d axes with G levels each produce $K = G^d$ environments. The direct row-multiplier implementation costs $O(BnK)$; the conditional-covariance alternative trades this for $O(nK^2 + K^3 + BK^2)$ work and $O(K^2)$ storage. Appendix B reports measured bootstrap timings and variability. These computational costs reinforce the design recommendation to keep stress axes scientifically focused or to use independent-pilot screening rather than an indiscriminately dense high-dimensional grid.

What is established. We give a coherent assurance target, transparent sampling and inference contracts, a standard finite-sample route, a justified fixed-family multiplier construction, corrected mechanism simulations, explicit reference uncertainty, and reproducible comparisons. The framework accepts any frozen pipeline yielding bounded losses, but the experiments cover a small number of tabular model classes. Interface generality is not empirical evidence for arbitrary neural architectures or deployment systems.

What remains open. The inferential components do not constitute a new optimality theorem. No minimax recovery rate, general safe exploration guarantee, or efficient time-uniform adaptive grid is proved. The continuous extension depends on external smoothness. Unknown shifts and causal transportability remain outside the certificate. Unbounded heavy-tailed losses and adversarial adaptation are not studied. Reference-risk uncertainty persists near the tolerance even with 20,000 rows. The primary simulation deliberately calibrates inference, and WDBC remains small and semisynthetic. The completed MIMIC-IV experiment adds a large credentialed EHR application with approximate temporal grouping, but it does not establish measured covariate or outcome shift, uses admission-table predictors from one institutional data source, and adds synthetic MCAR missingness. The permissive primary 0–1-loss specification certifies a trivial always-survive baseline; the secondary false-negative-risk specification is more discriminating, but it is an additional assurance target and is not a prospective clinical validation. These limitations still prevent a claim of general deployment readiness or clinical safety.

10 Conclusion

The proposed workflow connects a prespecified deployment family to an explicit, auditable set-valued risk report. Established finite-sample and bootstrap tools provide different routes to that report, with different assumptions and efficiency. The experiments show both the benefit of accounting for dependence and consequential Gaussian-approximation failures; exact methods remain the default for binary-loss certification in this work. The credentialed MIMIC-IV run demonstrates end-to-end execution on a large EHR cohort with approximate

temporal groups, native missingness, unequal group sizes, and class imbalance, without claiming that those group labels establish measured covariate or outcome shift. Its two risk specifications illustrate the scientific dependence of a certificate on the target: the permissive 0–1-risk tolerance certifies an always-survive classifier, whereas a secondary false-negative-risk target certifies only 3 of 60 eligible conditions, all for histogram gradient boosting. The appropriate conclusion is therefore a scoped assurance framework with reproducible synthetic, tabular, and EHR evidence, not a general claim of deployment readiness or clinical safety. Stronger clinical claims require prospectively chosen task-specific losses and tolerances, broader external validation, and evidence beyond the standard simultaneous-inference and smoothness arguments used here.

Data and Code Availability

The source code, canonical analysis notebook, tests, current-run CSVs, generated figures, manuscript sources, and completed aggregate MIMIC-IV benchmark outputs are publicly available at `https://github.com/dfmaale/sciguard-sar`. The archived software record is available at Zenodo: `https://doi.org/10.5281/zenodo.23241355`. The WDBC CSV and simulation surface are supplied directly. MIMIC-IV is credentialed and is not redistributed; the repository instead contains the frozen v3.1 protocol, cohort-flow/result/diagnostic tables, natural-missingness summaries, decision maps, and SHA-256 hashes of the local source tables used for the run. Patient-level MIMIC data, patient-level derivatives, and fitted clinical models are not included.

Appendix A. Additional Tables

Table 10: Paired mean ARR gain of multiplier over each comparator. MCSE uses paired replication differences. These efficiency comparisons must be read with coverage results.

n	ρ	Comparator	Paired ARR gain	MCSE
500	0.55	Bonferroni normal	0.009	0.0016
1000	0.10	Bonferroni normal	-0.000	0.0008
1000	0.55	Bonferroni normal	0.012	0.0016
1000	0.90	Bonferroni normal	0.066	0.0038
5000	0.55	Bonferroni normal	0.012	0.0017
500	0.55	Bonferroni exact	0.037	0.0029
1000	0.10	Bonferroni exact	0.020	0.0020
1000	0.55	Bonferroni exact	0.036	0.0029
1000	0.90	Bonferroni exact	0.092	0.0041
5000	0.55	Bonferroni exact	0.016	0.0019
500	0.55	Exact Holm	0.035	0.0029
1000	0.10	Exact Holm	0.011	0.0016
1000	0.55	Exact Holm	0.024	0.0023
1000	0.90	Exact Holm	0.080	0.0041
5000	0.55	Exact Holm	0.003	0.0010

Table 11: Equally spaced WDBC logistic-regression grid ablation.

Grid	Method	K	Certified $\tau = .08$	Fraction
4x4	Multiplier	16	10	0.625
4x4	Bonferroni exact	16	7	0.438
6x6	Multiplier	36	23	0.639
6x6	Bonferroni exact	36	11	0.306
10x10	Multiplier	100	69	0.690
10x10	Bonferroni exact	100	23	0.230

Appendix B. Computational Cost and Bootstrap Variability

Algorithm 1: SciGuard statistical assurance region

Input: Frozen predictive pipeline g; candidate environments $\mathcal{E}_n$; assurance sample; tolerance τ; confidence level $1-\alpha$; bootstrap size B.

Evaluate g across all environments and form the paired loss matrix $\mathbf{L} \in \mathbb{R}^{n \times K_n}$

Compute $\widehat{R}_k$ and $\widehat{\sigma}_k$ for all environments

for $b = 1, \ldots, B$ **do**

  Draw Gaussian multipliers $\xi_1^{(b)}, \ldots, \xi_n^{(b)}$

  Compute $T_b^* = \max_k(-Z_{kb}^*)$

Set $\widehat{c}_{1-\alpha}$ to the empirical $(1-\alpha)$ quantile of $\{T_b^*\}_{b=1}^B$

Compute simultaneous upper bounds U_k from (7); set $U_k = 1$ for zero empirical variance

return $\widehat{\mathcal{A}}_{\tau,\alpha}(g) = \{e_k : U_k \leq \tau\}$ *and the full upper-risk surface*

The direct multiplier sampler requires $O(BnK)$ operations. Sampling from its conditional Gaussian covariance instead costs $O(nK^2 + K^3 + BK^2)$ and stores a $K \times K$ covariance; it can be advantageous for moderate K. The code chooses that route when $K \leq \min(n, 2000)$ and retains a chunked row-multiplier route otherwise. Conditional distributional equivalence is tested with identical-column and independent-column constructions. Numerical agreement between samplers does not verify frequentist coverage.

The timing study holds the loss matrix fixed and repeats each bootstrap size under 25 independent seeds. Table 12 separates the number of observations, environments, and bootstrap draws. Timings depend on machine and numerical libraries and exclude loss generation/model fitting. The critical-value MCSE is the standard error of its mean across seeds, not a guaranteed quantile error bound. Per-seed critical values and certified counts are also exported so bootstrap variability is inspectable directly.

Appendix C. High-Dimensional Gaussian Approximation Note

For increasing K, the high-dimensional Gaussian-approximation results of Chernozhukov et al. (2013) provide sufficient regimes for bounded coordinates with uniformly positive variances; a familiar conservative regime is $\log^7(Kn)/n \to 0$ under the associated regularity constants. Lemma 1 establishes the independent-vector construction used here, while

Table 12: Conditional-covariance multiplier timing and variability.

n	K	B	Critical mean	Critical MCSE	Mean seconds
500	25	200	2.819	0.0188	0.0015
500	25	1000	2.795	0.0085	0.0021
500	25	5000	2.796	0.0052	0.0055
1000	25	200	2.806	0.0289	0.0018
1000	25	1000	2.793	0.0089	0.0025
1000	25	5000	2.794	0.0037	0.0055
1000	125	200	3.257	0.0227	0.0096
1000	125	1000	3.213	0.0097	0.0152
1000	125	5000	3.210	0.0035	0.0322

bounded losses and a uniform variance lower bound supply basic moment conditions. Later refinements improve Gaussian and bootstrap approximation rates (Chernozhukov et al., 2022). We do not assert their sharpest rates or optimality for the present grouped and stressed-data implementation, and none of these rates is used as a practical sample-size prescription. The finite-sample stress experiments instead assess approximation behavior directly.

Appendix D. Planning and Verification

Table 13: Conservative assurance-sample planning for a Hoeffding additive width at most the stated margin, with $\alpha = .05$. This is not a guarantee of a particular recall.

K	Margin	Sufficient n
25	0.02	7769
25	0.05	1243
25	0.10	311
100	0.02	9502
100	0.05	1521
100	0.10	381
500	0.02	11513
500	0.05	1843
500	0.10	461

Tests cover finite-sample binomial error by enumerating possible counts, the zero-error bound, empirical degeneracy, duplicated columns, covariance/row multiplier equivalence, unequal samples, overlapping and empty groups, reference boundaries, MAR row locality, ATC ties, and admission-aligned MIMIC cohort construction. A synthetic MIMIC fixture exercises the local pipeline and aggregate-output schema during tests and is discarded afterward; it is not treated as benchmark evidence. The credentialed v3.1 execution reported in Section 8 is stored separately as aggregate outputs with a frozen configuration and source-table hashes. The canonical notebook imports these same functions, records configuration and package versions, and generates manuscript inputs from completed runs rather than

hard-coded target values. All tables and figures in this manuscript are regenerated from the current corrected implementation with seed 20260927; rerunning the released notebook with the frozen configuration reproduces the reported outputs, including the secondary MIMIC false-negative tables and assurance map. Historical draft outputs are superseded by this implementation, and no reported result was selected by changing the seed.

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